

8. (a) **Complete the table** by writing in **true** or **false** against each of the following pairs of statements about antibiotics and vaccines. [2]

Statements	True or False
Antibiotics are produced from fungi. Vaccines are produced from dead or non-active pathogens.	
Antibiotics are used to prevent infection. Vaccines are used to treat infection.	
Antibiotics can cure infection caused by bacteria. Vaccines cannot cure infection caused by bacteria.	

(b) Smallpox was an infectious disease caused by a virus. An estimated 300 million people worldwide died from smallpox during the 20th Century. The last naturally occurring case was diagnosed in October 1977 and the World Health Organization certified the global eradication of the disease in 1980.

The global eradication was considered to have occurred because the smallpox vaccine was very effective and the programme of vaccination was thorough and worldwide.

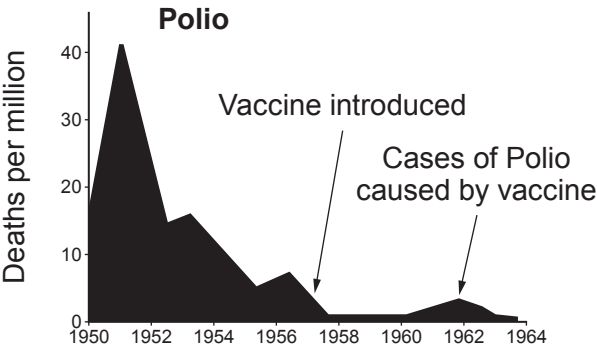
However, some authorities dispute the claim that eradication was due to vaccination. They stated that deaths from smallpox were already decreasing in unvaccinated populations and that only 10% of the World’s population were vaccinated.



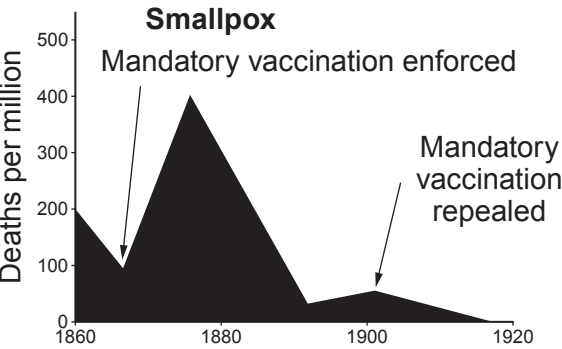
Examiner
only

Graphs 8.1, 8.2, 8.3 and 8.4 show data for four diseases that caused deaths in the UK in the 19th and 20th Centuries. They also show the dates at which vaccines for three of these diseases were introduced.

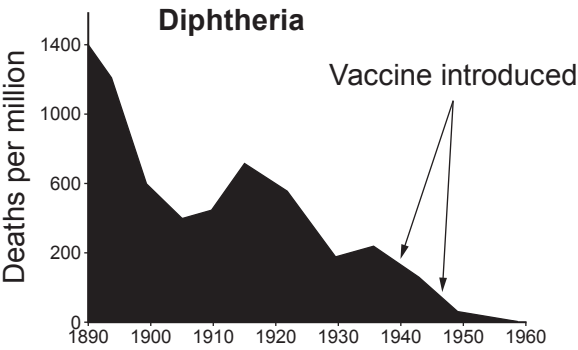
Graph 8.1



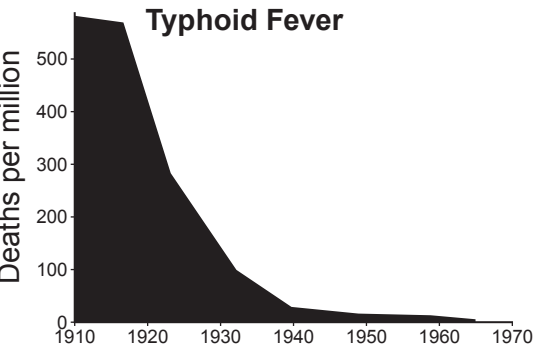
Graph 8.2



Graph 8.3



Graph 8.4



Note: **Mandatory vaccination enforced** means that by law people had to be vaccinated against smallpox.
Mandatory vaccination repealed means that people were no longer legally required to be vaccinated against smallpox.

Using **only** the information in **Graphs 8.1, 8.2, 8.3 and 8.4**, explain how they could be used as evidence both **for** and **against** the use of vaccination. [3]

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END OF PAPER

